

2.8 Predicate Logic

Predicate logic is an extension of propositional logic. It allows us to express more detailed statements about objects and their properties. We can talk about individuals, groups, and relationships between them.

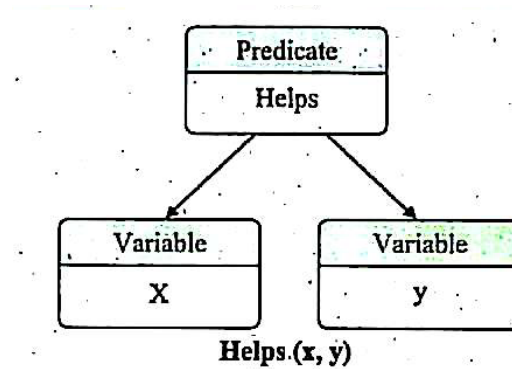


Figure 2.7: Predicate Logic Structure

Introduction to Predicate Logic

Predicate logic studies statements that contain variables and predicates.

What is Predicate?

Definition: A **predicate** is a statement that can be true or false depending on the values of the variables.

Example: “x is greater than 5” is a predicate because it depends on x.

Why We Use Predicate Logic?

- It allows expressing statements like “All students are hardworking” or “Some books are interesting.”
- It is used to represent knowledge formally.

Difference Between Propositional and Predicate Logic

Feature	Propositional Logic	Predicate Logic
Nature	Propositional logic deals with simple statements that are either true or false.	It uses predicates and variables to express detailed statements.
Expressiveness	It is less expressive.	Predicate, logic is more expressive and powerful
Ability	It cannot describe properties of objects or relationships between them.	It can describe describe properties of objects or relationships between them.

Flexibility	Propositional logic is easier to understand but less flexible.	It is more flexible. It allows statements about many objects and conditions.
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Example Comparison:

- Propositional Logic:
 - A = "Ali is Tall"
 - Cannot see inside(A)
 - Limited
- Predicate Logic:
 - Tall(Ali)
 - Can see subject + property Tall
 - Much more expressive)

Predicates and Variables

- **Predicates** describe properties or relationships of objects. Example: isEven(x) is true if x is an even number.
- **Variables** represent objects or values in the domain of discourse. Example: "For all x, $x > 0$ "

Predicates and variables together form predicate logic expressions and help express complex ideas clearly.

Short Questions

1. What is predicate logic?
2. How is predicate logic different from propositional logic?
3. What is a predicate? Give an example.
4. What is the role of variables in predicate logic?
5. Why is predicate logic more powerful than propositional logic?
6. Give one example of a statement in propositional logic and its equivalent in predicate logic.
7. What can predicate logic express that propositional logic cannot?

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